

單選 50 題 (每題 2 分共 100 分) 答錯不倒扣

1. How many protons, neutrons, and electrons are present in an atom of the sulfur-31 isotope? (p = proton, n = neutron, e = electron)
(A) 15 p, 16 n, 15 e (B) 16 p, 15 n, 16 e (C) 15 p, 16 n, 16 e
(D) 15 p, 15 n, 16 e (E) 16 p, 16 n, 15 e
2. What is the empirical formula of manganosite, a compound consisting of manganese-55 and oxygen-16, if 77% of its mass is attributed to manganese?
(A) MnO (B) Mn₂O₃ (C) MnO₂ (D) Mn₂O₂ (E) Mn₃O₂
3. The oxidation number of S in K₂SO₄ is
(A) +6 (B) +4 (C) +2 (D) -2 (E) -4
4. What are the predicted products of the single replacement reaction between solid iron (Fe) and aqueous copper(II) sulfate?
(A) Cu(s) + FeSO₄(aq) (B) Fe(s) + Cu(s) + SO₄(aq) (C) CuS(s) + Fe₂SO₄(aq)
(D) FeCuSO₄(aq) (E) FeO(s) + CuSO₃(aq)
5. Calculate the density of SF₆ gas at a temperature of 27°C and a pressure of 0.500 atm, expressed in grams per liter (g/L).
(A) 3.38 g/L (B) 2.24 g/L (C) 2.96 g/L (D) 22.4 g/L (E) 33.8 g/L
6. Among these compounds, which one is most likely to exhibit covalent bonding?
(A) Rb₂S (B) SrCl₂ (C) CS₂ (D) CaO (E) MgI₂
7. Among these pairs of atoms, which one would likely form the most polar bond?
(A) B—C (B) C—N (C) C—O (D) Si—O (E) C—C
8. Among these compounds, which one is most likely to exhibit ionic bonding?
(A) NCl₃ (B) BaCl₂ (C) CO (D) SO₂ (E) SF₄
9. Which of the following molecules possesses tetrahedral geometry?
(A) XeF₄ (B) BF₃ (C) AsF₅ (D) CF₄ (E) NH₃
10. How many atoms are present in a body-centered cubic unit cell?
(A) 1 (B) 2 (C) 3 (D) 4 (E) 8
11. In a face-centered cubic lattice of a metal, each atom is surrounded by how many nearest neighbors (atoms in contact)?
(A) 4 (B) 6 (C) 8 (D) 10 (E) 12
12. Given that the molar heats of sublimation and fusion of iodine are 62.3 kJ/mol and

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- 15.3 kJ/mol, respectively, calculate the molar heat of vaporization of liquid iodine.
(A) 77.6 kJ/mol (B) 47.0 kJ/mol (C) -47.0 kJ/mol (D) -77.6 kJ/mol (E) 7.76 kJ/mol
13. Which of the following statements is false according to Raoult's law?
(A) The vapor pressure of a solvent over a solution decreases as its mole fraction increases. (B) The solubility of a gas increases as the temperature decreases. (C) The vapor pressure of a solvent over a solution is less than that of pure solvent. (D) The greater the pressure of a gas over a solution the greater its solubility. (E) Ionic solutes dissociate in solution causing an enhancement of all colligative properties.
14. Among these 0.10 M solutions, which one will have the lowest pH?
(A) HCN (B) HNO₃ (C) NaCl (D) H₂CO₃ (E) NaOH
15. Arrange the acids HOCl, HClO₃, and HClO₂ in order of increasing acidity.
(A) HOCl < HClO₃ < HClO₂ (B) HOCl < HClO₂ < HClO₃ (C) HClO₂ < HOCl < HClO₃ (D) HClO₃ < HOCl < HClO₂ (E) HClO₃ < HClO₂ < HOCl
16. At constant T and P, a negative sign for ΔG indicates that
(A) the reaction is exothermic. (B) the reaction is endothermic. (C) the reaction is fast. (D) the reaction is spontaneous. (E) ΔS must be > 0.
17. For the reaction $\text{H}_2(\text{g}) + \text{S}(\text{s}) \rightarrow \text{H}_2\text{S}(\text{g})$, $\Delta H^\circ = -20.2 \text{ kJ/mol}$ and $\Delta S^\circ = +43.1 \text{ J/K} \cdot \text{mol}$. Which of these statements is true?
(A) The reaction is only spontaneous at low temperatures. (B) The reaction is spontaneous at all temperatures. (C) ΔG° becomes less favorable as temperature increases. (D) The reaction is spontaneous only at high temperatures. (E) The reaction is at equilibrium at 25°C under standard conditions.
18. In the complex ion $[\text{Cr}(\text{C}_2\text{O}_4)_2(\text{H}_2\text{O})_2]^-$, the oxidation number of Cr is
(A) +1 (B) +2 (C) +3 (D) +4 (E) +5
19. How many unpaired electrons are there in the complex ion $[\text{Mn}(\text{CN})_6]^{3-}$?
(A) 0 (B) 1 (C) 2 (D) 3 (E) 4
20. Which of the following species are radicals?
(A) CH₂O (B) HCN (C) HClO (D) ClONO₂ (E) ClO
21. Which of the compounds listed below has the weakest hydrogen bonds?
(A) CH₄ (B) H₂O (C) SiH₄ (D) HF (E) H₂S

22. Why is the bond dissociation energy of C—C greater than that of C—H?
(A) due to enhanced sigma bond overlap. (B) because the bond is electrostatically stronger. (C) because multiple bonds are always stronger than single bonds. (D) due to the decreased bond dipole. (E) It is not; the dissociation energy is greater for C—H.
23. All of the following have a linear shape except for which one
(A) IF_2^- (B) CS_2 (C) XeF_2 (D) I_3^- (E) NH_2^-
24. If 2.0 moles of an ideal gas at STP are subjected to a new pressure of 24.7 kPa, what will be the volume of the gas?
(A) 8.20 L (B) 22.4 L (C) 11.2 L (D) 184 L (E) 91.9 L
25. Consider the following reaction: $4\text{NaO}_2(\text{s}) + 2\text{CO}_2(\text{g}) \rightarrow 2\text{Na}_2\text{CO}_3(\text{s}) + 3\text{O}_2(\text{g})$. How many moles of NaO_2 are required to react with 75.0 L of CO_2 at STP?
(A) 0.15 (B) 1.67 (C) 3.35 (D) 6.70 (E) 13.4
26. What mass of propane needs to be burned to provide 2580 kJ of heat? The standard enthalpy of combustion of propane at 298 K is $-2220 \text{ kJ} \cdot \text{mol}^{-1}$.
(A) 25.6 g (B) 51.2 g (C) 102 g (D) 75.9 g (E) 37.9 g
27. What type of process is the formation of solid magnesium chloride from gaseous magnesium ions?
(A) a state function (B) an adiabatic process (C) an endothermic process (D) an exothermic process (E) not enough information is given to permit an answer
28. Which one is always true for a spontaneous process at constant temperature?
(A) $\Delta S_{\text{system}} + \Delta S_{\text{surroundings}} = q/T$ (B) $\Delta S > 0$ (C) $\Delta S = q/T$ (D) $\Delta S_{\text{system}} + \Delta S_{\text{surroundings}} > 0$ (E) $\Delta S < q/T$
29. The sublimation of solid carbon dioxide is a spontaneous process. Predict the sign (+, -, or 0) of ΔG_r° , ΔH_r° , and ΔS_r° , respectively.
(A) -, 0, + (B) -, -, - (C) -, +, + (D) 0, +, + (E) -, +, -
30. Which of the following is the weakest acid?
(A) HNO_3 (B) HBr (C) HCl (D) HF (E) HI
31. Which of the following is not a Lewis base
(A) OH^- (B) H_2O (C) SO_3 (D) Br^- (E) NH_3
32. Which of the following has the highest pK_a value?
(A) HClO (B) HClO_3 (C) HClO_4 (D) HBrO (E) HIO

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33. What is the pH of an aqueous solution containing 0.12 M $\text{C}_6\text{H}_5\text{NH}_2$ ($K_b = 4.3 \times 10^{-10}$) and 0.018 M $\text{C}_6\text{H}_5\text{NH}_3\text{Cl}$?
(A) 5.46 (B) 10.19 (C) 4.63 (D) 3.81 (E) 8.54
34. Which of the following is the strongest reducing agent?
(A) Fe^{2+} (B) Cr^{2+} (C) F^- (D) H_2 (E) Co^{2+}
35. A reaction that has a very low activation energy exhibits which of the following characteristics?
(A) The rate is very sensitive to temperature. (B) The rate does not change significantly with temperature. (C) It must be second order. (D) It produces a curved Arrhenius plot. (E) It must be first order.
36. Which of the following statements regarding the effect of a catalyst on a reaction is not true?
(A) A catalyst has no effect on the equilibrium composition of the reaction.
(B) A catalyst provides a lower activation energy for the reaction. (C) When a reaction is catalyzed, both forward and reverse reactions are accelerated. (D) A catalyst speeds up a reaction by providing an alternate pathway for the reaction.
(E) When a catalyst speeds up a reaction, the rate law remains the same.
37. The chemical formula for diborane is
(A) BH_3 (B) B_2H_6 (C) B_2H_4 (D) B_2H_2 (E) B_2H
38. The geometry of PCl_4^+ is
(A) T-shaped. (B) tetrahedral. (C) seesaw. (D) trigonal bipyramidal. (E) trigonal pyramidal.
39. Which of the following is considered unstable?
(A) HFO (B) HBrO (C) HClO (D) HI (E) HF
40. Which of the following is not an example of a redox reaction?
(A) $3\text{Mg} + \text{N}_2 \rightarrow \text{Mg}_3\text{N}_2$
(B) $2\text{P}^{3-} + 6\text{H}_2\text{O} \rightarrow 2\text{PH}_3 + 6\text{OH}^-$
(C) $2\text{NaN}_3 \rightarrow 2\text{Na} + \text{N}_2$
(D) $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$
(E) $\text{N}_2\text{H}_4 + \text{O}_2 \rightarrow \text{N}_2 + 2\text{H}_2\text{O}$
41. How many unpaired electrons can be predicted for a tetrahedral iron(II) complex?
(A) 1 (B) 2 (C) 3 (D) 4 (E) 5
42. Elements with an odd number of protons and neutrons
(A) are usually stable. (B) are usually unstable. (C) always have a proton-to-

neutron ratio of about 1. (D) never undergo α emission. (E) have a magic number of protons or neutrons.

43. When a β particle is emitted, the mass number

(A) decreases by 4. (B) increases by 1. (C) decreases by 2. (D) decreases by 1. (E) does not change.

44. Which of the following is a secondary alcohol?

(A) $\text{CH}_3\text{CH}_2\text{OH}$ (B) $\text{C}_6\text{H}_5\text{CH}_2\text{OH}$ (C) $\text{C}(\text{CH}_3)_3\text{OH}$ (D) $\text{CH}_3(\text{CH}_2)_3\text{OH}$
(E) $\text{CH}(\text{CH}_3)_2\text{OH}$

45. The $-\text{OH}$ group occurs in

(A) aldehydes. (B) amides. (C) carboxylic acids. (D) ketones. (E) esters.

46. Which of the following compounds lacks the carbonyl group?

(A) amides. (B) ketones. (C) carboxylic acids. (D) aldehydes. (E) phenols.

47. Which of the following terms refers to a peptide bond?

(A) $-\text{C}(\text{O})\text{O}-$ (B) $-\text{CONH}-$ (C) $-\text{C}(\text{O})\text{OC}(\text{O})-$ (D) $-\text{C}=\text{N}-$ (E) $-\text{OP}(\text{O})_2\text{O}-$

48. How many structural isomers can be formed for hexane?

(A) 4 (B) 5 (C) 6 (D) 7 (E) 8

49. Consider a molecule with a molecular formula of $\text{C}_{10}\text{H}_{16}$. How many rings and π bonds are present in the molecule in total?

(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

50. Which of the following compounds exhibit a trans configuration?



I

II

III

IV

(A) I and IV (B) II and III (C) I and II (D) I and III (E) II and IV